

Pinned group summary: $SL(n=4)$

Pinning-Sympy automated output

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1 Basic group attributes

Group	SL(n=4)
Matrix size	4×4
Rank	3
Defining equations	<i>DefiningEquationPlaceholder</i>

1.1 Lie algebra

The defining equations for the Lie algebra are:

LieAlgebraEquationPlaceholder

A generic element of the Lie algebra is:

$$\begin{bmatrix} x_{00} & x_{01} & x_{02} & & x_{03} \\ x_{10} & x_{11} & x_{12} & & x_{13} \\ x_{20} & x_{21} & x_{22} & & x_{23} \\ x_{30} & x_{31} & x_{32} & -x_{00} - x_{11} - x_{22} & \end{bmatrix}$$

1.2 Torus

The torus has rank 3. A generic element of the torus is

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 \\ 0 & 0 & t_2 & 0 \\ 0 & 0 & 0 & \frac{1}{t_0 t_1 t_2} \end{bmatrix}$$

The rows of the matrix below are trivial characters:

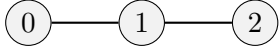
$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Dynkin type	A_3
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	12

2.1 Dynkin diagram

The root system is irreducible and has Dynkin diagram:



2.2 Cartan matrix

The root system is irreducible and has Cartan matrix:

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

2.3 Roots

Root	Simple	Sign	Multipliable
$(-1, 0, 0, 1)$	No	—	No
$(-1, 0, 1, 0)$	No	—	No
$(-1, 1, 0, 0)$	No	—	No
$(0, -1, 0, 1)$	No	—	No
$(0, -1, 1, 0)$	No	—	No
$(0, 0, -1, 1)$	No	—	No
$(0, 0, 1, -1)$	Yes	+	No
$(0, 1, -1, 0)$	Yes	+	No
$(0, 1, 0, -1)$	No	+	No
$(1, -1, 0, 0)$	Yes	+	No
$(1, 0, -1, 0)$	No	+	No
$(1, 0, 0, -1)$	No	+	No

2.4 Coroots

Root	Coroot
$(-1, 0, 0, 1)$	$(-1, 0, 0, 1)$
$(-1, 0, 1, 0)$	$(-1, 0, 1, 0)$
$(-1, 1, 0, 0)$	$(-1, 1, 0, 0)$
$(0, -1, 0, 1)$	$(0, -1, 0, 1)$
$(0, -1, 1, 0)$	$(0, -1, 1, 0)$
$(0, 0, -1, 1)$	$(0, 0, -1, 1)$
$(0, 0, 1, -1)$	$(0, 0, 1, -1)$
$(0, 1, -1, 0)$	$(0, 1, -1, 0)$
$(0, 1, 0, -1)$	$(0, 1, 0, -1)$
$(1, -1, 0, 0)$	$(1, -1, 0, 0)$
$(1, 0, -1, 0)$	$(1, 0, -1, 0)$
$(1, 0, 0, -1)$	$(1, 0, 0, -1)$

2.5 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$.

α	β	Pairs (i, j)
$(-1, 0, 0, 1)$	$(0, 0, 1, -1)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(0, 1, 0, -1)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(1, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(1, 0, -1, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(0, 0, -1, 1)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(0, 1, -1, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(1, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(1, 0, 0, -1)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(0, -1, 0, 1)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(0, -1, 1, 0)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(1, 0, -1, 0)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, -1, 0, 1)$	$(0, 0, 1, -1)$	$(1, 1)$
$(0, -1, 0, 1)$	$(0, 1, -1, 0)$	$(1, 1)$
$(0, -1, 0, 1)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(0, 0, -1, 1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(0, 1, 0, -1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(1, 0, -1, 0)$	$(1, 1)$
$(0, 0, -1, 1)$	$(0, 1, 0, -1)$	$(1, 1)$
$(0, 0, -1, 1)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, 0, 1, -1)$	$(0, 1, -1, 0)$	$(1, 1)$
$(0, 0, 1, -1)$	$(1, 0, -1, 0)$	$(1, 1)$
$(0, 1, -1, 0)$	$(1, -1, 0, 0)$	$(1, 1)$
$(0, 1, 0, -1)$	$(1, -1, 0, 0)$	$(1, 1)$

3 Root spaces

3.1 Dimensions

The table below lists each root along with the dimension of the associated root space (and root subgroup).

Root (α)	Dimension of root space (d_α)
$(-1, 0, 0, 1)$	1
$(-1, 0, 1, 0)$	1
$(-1, 1, 0, 0)$	1
$(0, -1, 0, 1)$	1
$(0, -1, 1, 0)$	1
$(0, 0, -1, 1)$	1
$(0, 0, 1, -1)$	1
$(0, 1, -1, 0)$	1
$(0, 1, 0, -1)$	1
$(1, -1, 0, 0)$	1
$(1, 0, -1, 0)$	1
$(1, 0, 0, -1)$	1

3.2 Root space table

The table below lists each root along with a generic element of the associated root space (inside the Lie algebra).
Include a definition here? INCOMPLETE

Root (α)	Generic element of root space			
$(-1, 0, 0, 1)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ u_0 & 0 & 0 & 0 \end{bmatrix}$		
$(-1, 0, 1, 0)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ u_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		
$(-1, 1, 0, 0)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ u_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		
$(0, -1, 0, 1)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & u_0 & 0 & 0 \end{bmatrix}$		
$(0, -1, 1, 0)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & u_0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		
$(0, 0, -1, 1)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & u_0 & 0 \end{bmatrix}$		
$(0, 0, 1, -1)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & u_0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		
$(0, 1, -1, 0)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & u_0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		
$(0, 1, 0, -1)$		$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & u_0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		
$(1, -1, 0, 0)$		$\begin{bmatrix} 0 & u_0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		
$(1, 0, -1, 0)$		$\begin{bmatrix} 0 & 0 & u_0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		
$(1, 0, 0, -1)$		$\begin{bmatrix} 0 & 0 & 0 & u_0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$		

4 Root subgroups

4.1 Root subgroup table

The table below lists each root along with a generic element of the associated root subgroup.

RootSubgroupTablePlaceholder

4.2 Homomorphism property

Background theoretical info on homomorphism/pseudo-homomorphism property INCOMPLETE

HomomorphismDefectTablePlaceholder

Below is an explicit formulation of the homomorphism/pseudo-homomorphism identities.

HomomorphismDefectIdentitiesPlaceholder

5 Commutators

Background theoretical info on commutator relation INCOMPLETE

Because of its relevance to commutators, we repeat here the table of linear combinations of roots.

α	β	Pairs (i, j)
$(-1, 0, 0, 1)$	$(0, 0, 1, -1)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(0, 1, 0, -1)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(1, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 1)$	$(1, 0, -1, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(0, 0, -1, 1)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(0, 1, -1, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(1, -1, 0, 0)$	$(1, 1)$
$(-1, 0, 1, 0)$	$(1, 0, 0, -1)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(0, -1, 0, 1)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(0, -1, 1, 0)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(1, 0, -1, 0)$	$(1, 1)$
$(-1, 1, 0, 0)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, -1, 0, 1)$	$(0, 0, 1, -1)$	$(1, 1)$
$(0, -1, 0, 1)$	$(0, 1, -1, 0)$	$(1, 1)$
$(0, -1, 0, 1)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(0, 0, -1, 1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(0, 1, 0, -1)$	$(1, 1)$
$(0, -1, 1, 0)$	$(1, 0, -1, 0)$	$(1, 1)$
$(0, 0, -1, 1)$	$(0, 1, 0, -1)$	$(1, 1)$
$(0, 0, -1, 1)$	$(1, 0, 0, -1)$	$(1, 1)$
$(0, 0, 1, -1)$	$(0, 1, -1, 0)$	$(1, 1)$
$(0, 0, 1, -1)$	$(1, 0, -1, 0)$	$(1, 1)$
$(0, 1, -1, 0)$	$(1, -1, 0, 0)$	$(1, 1)$
$(0, 1, 0, -1)$	$(1, -1, 0, 0)$	$(1, 1)$

5.1 Commutator coefficient table

The table below gives the coefficients $N_{ij}^{\alpha\beta}$.

CommutatorCoefficientTablePlaceholder

5.2 Commutator identities

Below is an explicit formulation of the commutator formula identities.

CommutatorIdentitiesPlaceholder

6 Weyl group

Background theoretical info on Weyl group, especially as realized inside the group, including info on conjugation between root subgroups INCOMPLETE

6.1 Weyl group elements

The table below lists roots along with an associated Weyl group element.

WeylGroupPlaceholder

6.2 Weyl group conjugation coefficients

The table below lists roots along with the coefficients obtain in performing conjugation by an associated Weyl element.

WeylConjugationCoefficientPlaceholder

6.3 Weyl group conjugation identities

Below is an explicit formulation of the Weyl group conjugation identities.

WeylConjugationIdentitiesPlaceholder